

Assignment #2

Shown in the pictures below is the expected result after completing the this assignment.

1. Within codes**8490, create two new java files `A2Shapes**.java` and `Assignment2**.java`, where ** is your initials.
2. The simulated wind turbine of A1 has two rotational components, i.e., the yaw drive and the blade rotor. Code your `A2Shapes**.java` file to introduce rotational behavior to the two objects with new classes modified from their previous classes in `BaseShapes**.java`. Afterwards, code your `Assignment2**.java` file to render the virtual world of a wind turbine with rotational behavior.
 - (a) The sphere-shaped yaw drive rotates continuously around its Y-axis counterclockwise at a speed of 20,000 milliseconds per round rotation.
 - (b) The sphere-shaped blade rotor rotates continuously around its Z-axis clockwise at a speed of 5,000 milliseconds per round rotation.
3. In addition, introduce the following changes to the virtual world.
 - (a) Replace the white base with a self-define square box that is in the same dimension as the original base, but its six surfaces are set to different colors of your choice.
 - (b) Make the new base half transparent and keep it stationary.
 - (c) Use texture mapping to wrap the two spheres with an image of your choice so that their rotations become easier to observe.
 - (d) Identify your work by updating the string label.
 - (e) Make your program easy to comprehend by adding adequate amount of comments.
4. Locate the folder that contains all files of your project `Comp8490`; produce a zip file of your project folder; and submit it online before due.

